

Exploring the AKS cluster – Azure infrastructure side



- **Managed Identity (MI)**

- With the help of a managed identity feature in Azure, resources can authenticate to other Azure services without the user having to deal with managing the authentication credentials
- This is a special type of service principal that provides automatic credential rotation, so you don't need to worry about credentials or their expiration
- Two types of managed identities exist:
 - System-assigned: tied to the lifecycle of an Azure resource and can be associated with a single Azure resource
 - User-assigned: standalone Azure resource with its own lifecycle that can be associated with multiple Azure resources
- In AKS, to manage cluster resources like load balancer, AKS managed public IP and others, an identity is used. This kind of identity can be a managed identity, and by default, when you create an AKS cluster, unless you create a user-assigned managed identity, a system-assigned managed identity gets automatically created, and this is managed by the Azure platform
- The managed identity in the infrastructure resource group is a user-assigned managed identity, and its main goal is to authenticate with the Azure Container Registry (ACR). This is also called Kubelet identity